

FIT5196 Data Wrangling

Assessment 2 Data Cleansing, Audit and Transformation

Semester	Semester 2, 2026
Weight	20% of the unit grade, marked out of 20
Mode	Group assessment; continue with your allocated project group
Due	23:55, 23 October 2026
Submission	One group ZIP through the Moodle A2 submission activity

Your group must investigate three order datasets, correct dirty values, recover missing values, remove delivery-charge outliers and demonstrate that the outputs are reliable. Then compare transformations for customer profiling using your cleaned orders. Submit the corrected CSVs, a reproducible notebook and Python source, and a field-level audit log. The separate A2 marking rubric explains the allocation of all 20 marks and how results and methods are evaluated.

1 Project context

A new staff member has prepared three incoming order extracts for a Melbourne technology retailer. One contains dirty values, one contains missing values, and one contains delivery-charge outliers. Your team must investigate and remediate these extracts before they enter the retailer's analytical system. The data is fictional.

This is the current-period stage of the project. A1 describes historical activity; A2 contains different, later orders for the same business universe. Historical prices are not necessarily current prices. All evidence needed to solve A2 is supplied in the A2 package and the rules below. **Recreating your A1 outputs is not an A2 requirement; no extra marks will be given for doing so.**

A2 assesses data cleansing, auditing and purpose-specific transformation of the cleaned retail data. No presentation, dashboard or separate suburb dataset is required for this A2.

By completing A2, you should be able to diagnose data-quality problems, select evidence-based repairs, estimate missing values, identify conditional outliers, preserve valid data, document and validate a reproducible workflow, and justify data transformations for a stated analytical purpose.

2 Files supplied and data structure

Use the ZIP allocated to your existing group number, `GroupNNN_A2.zip`, where `NNN` has three digits, for example `Group007`. Check this identifier before starting. Each group's work is compared with that group's reference answers.

Supplied file	Purpose
<code>GroupNNN_dirty_data.csv</code>	500 orders requiring investigation for dirty values
<code>GroupNNN_missing_data.csv</code>	500 orders with some values omitted
<code>GroupNNN_outlier_data.csv</code>	500 orders requiring delivery-charge outlier detection
<code>warehouses.csv</code>	Warehouse names, latitude and longitude
<code>input_manifest.csv</code>	Input filenames, row counts, column counts and grain
<code>resource_contract.json</code>	Frozen VADER resource location, checksum, NLTK version and sentiment threshold
<code>resources/nltk_data/sentiment/vader_lexicon.zip</code>	The VADER lexicon to use locally
<code>README.md</code>	The public conventions for your package
<code>A2_manifest.json</code>	Group identifier and checksums of the supplied files

Each order CSV has the following **16 columns in this order**. The grain is one row per order. Order IDs are unique within and across the three input batches; the files contain 1,500 distinct orders, not three versions of the same orders.

Column	Meaning and required interpretation
<code>order_id</code>	Stable order identifier; retain exactly and use for matching records
<code>customer_id</code>	Stable customer identifier; customers may have multiple orders

Column	Meaning and required interpretation
date	Order date; standardised output format is YYYY-MM-DD
nearest_warehouse	Name of the warehouse nearest to the customer, from warehouses.csv
shopping_cart	JSON array of objects with product and positive integer quantity keys
order_price	Sum of item quantities times current unit prices, before coupon discount and delivery; AUD
delivery_charges	Delivery charge in AUD, governed by the seasonal relationship below
customer_lat	Customer latitude in decimal degrees
customer_long	Customer longitude in decimal degrees
coupon_discount	Percentage discount on order_price; for example, 10 means 10%
order_total	Discounted item price plus delivery charge, in AUD
season	Australian season: Summer, Autumn, Winter or Spring
is_expedited_delivery	Boolean indicating expedited delivery
distance_to_nearest_warehouse	Haversine distance to the nearest warehouse, in kilometres
latest_customer_review	Supplied English prior-review text, or the exact sentinel No review submitted
is_happy_customer	Boolean derived from the supplied review using the published VADER rule

The reference file has columns warehouse, latitude, longitude and can be treated as correct. It describes the three warehouses serving Melbourne.

Batch guarantees

- A dirty-data row contains at most one intentionally incorrect field. A row may already be correct. The published evidence permits a defensible repair.
- The missing-data batch contains missing values only. Its other values are correct. Missing values are represented by empty CSV fields.
- The outlier batch contains anomalous `delivery_charges` only. It has no missing values or other intentionally corrupted fields.
- `order_id`, `customer_id`, `shopping_cart`, `customer_long`, `coupon_discount` and `latest_customer_review` are complete and error-free in all three batches. Preserve their information.
- An inconsistency between two fields does not prove both are wrong. Use independent evidence and the batch guarantees to identify the affected field.
- Do not assume a fixed percentage of bad records or infer corrections from row positions or patterns in order IDs. IDs are for matching and tracing.

Keep the files separate for diagnosis and final outputs. You may pool suitable evidence across your own A2 batches when justified, while retaining source-file identity and excluding missing or unreliable training values. Another group's data is not a training or answer source. Task 3 combines your three cleaned outputs into a separate analysis copy; it does not merge the required CSV deliverables into one file.

3 Published business and calculation rules

3.1 Shopping carts and current prices

Parse `shopping_cart` with a JSON parser. For example, a cell may represent `[{"product": "Example Item", "quantity": 2}]`. Doubled quotation marks in the raw CSV are standard CSV escaping; reading the CSV normally restores the JSON string. Do not use `eval` to parse it.

Within your A2 package, a product has one current unit price across the three batches. Infer these prices from sufficient reliable shopping-cart and `order_price` equations. Explain the equations, evidence selection and how you checked the inferred prices. Libraries such as NumPy may be used. A1 catalogue prices and another group's inferred prices are not authoritative for A2.

3.2 Discounts totals and rounding

Use this sequence:

1. Multiply each cart quantity by its inferred current unit price, round each item amount to two decimal places, then sum and round to two decimal places to obtain `order_price`.
2. Apply `coupon_discount / 100` to the item price only.

3. Add `delivery_charges` after the discount.
4. Round the final `order_total` to two decimal places. Do not separately round the discounted subtotal before adding delivery.

`order_total = round(order_price * (1 - coupon_discount / 100) + delivery_charges, 2)`

For an illustrative order with an item price of AUD \$100, a 10% coupon and an AUD \$8 delivery charge, the total is AUD \$98. The coupon does not discount delivery. There is no separate tax calculation or `tax_amount` output in A2; do not add GST again or copy additional columns from A1.

Keep valid source values. A model-estimated replacement charge may differ slightly because of rounding; that alone does not justify overwriting an otherwise valid total. Explain small estimation differences in your checks.

3.3 Dates seasons and location

Interpret `dates` as calendar dates. For a slash-formatted `day/month/year` value, use day first; a year-first form retains its year-first interpretation. Return all dates as `YYYY-MM-DD`.

Season	Months
Summer	December, January, February
Autumn	March, April, May
Winter	June, July, August
Spring	September, October, November

Calculate haversine distance from the customer to each supplied warehouse, using the published coordinates and an Earth radius of **6,371 km**. Select the smallest distance. This is straight-line great-circle distance, not a road-network route; graph files and a shortest-path algorithm are not needed. Inspect whether coordinates are consistent with Melbourne before relying on their distances.

Retain coordinate precision. Return reconstructed distances to four decimal places in kilometres and reconstructed monetary values to two decimal places.

3.4 Prior reviews and happiness

Apply NLTK VADER to `latest_customer_review` exactly as supplied. The field is already English-only and suitable for this task. Do not translate it, remove punctuation, rewrite it, apply A1 Latin-script filtering, or replace it with a newly generated review before scoring.

- Use the package-local frozen lexicon and the NLTK version stated in `resource_contract.json`. Verify the resource checksum and configure NLTK to load that local resource. A substitute downloaded lexicon is not the contract.
- Set `is_happy_customer` to `True` when the VADER compound score is **greater than or equal to 0.05**, and `False` otherwise.
- Treat the exact text `No review submitted` as a documented no-prior-review sentinel, with `is_happy_customer = True`. It is not a missing value to impute.

The supplied review is the latest eligible prior English review for that customer, written before the current order. You do not have to recreate its selection from A1 or establish its language independently.

3.5 Seasonal delivery models

Delivery charge follows **a separate linear relationship for each of the four seasons**. The predictors are `distance_to_nearest_warehouse`, `is_expedited_delivery` and `is_happy_customer`, with an intercept. **Encode the boolean predictors consistently as 0/1**; using the complementary unhappy indicator is equivalent when explained and fitted consistently. Infer model coefficients from your supplied data; they are not given as an answer list.

Use reliable complete observations, or values you have already repaired and validated. Document your training and evaluation selections. **Evaluate the relationship on suitable held-out observations, with a reproducible split, and report per-season sample sizes, R-squared and an absolute-error measure**. Clean data supports a very strong relationship, normally R-squared at least 0.99; investigate a poor fit. R-squared alone does not prove the model or repairs are correct, and an in-sample score is not held-out evidence.

Distinguish a large charge from a large residual, the difference between an observed charge and the charge expected from its predictors. A valid delivery may be expensive; an erroneous charge may lie within the ordinary raw-charge range. Select and justify your residual decision rule. The numerical comparison tolerances in Section 5 are marking tolerances, not a required outlier threshold.

4 Required work

Task 1 Data cleansing and explanation (15 marks)

Investigate the data first. In the solution notebook, **show graphical and/or non-graphical EDA that supports diagnosis: schema and type checks, missingness, distributions, categorical values, key checks and relevant relationships**. Explain what the evidence suggests and which observations are suitable for subsequent calculations or models.

Repair the dirty batch. Detect inconsistent or incorrectly represented values using the business rules and evidence. Identify which field is wrong, correct only the affected information, and retain all 500 order IDs. Where several checks fail because of one corrupted field, explain the dependency. Do not repeatedly modify other fields to force agreement with the bad value.

Recover the missing batch. Identify empty fields and choose a treatment appropriate to each field. Prefer calculable relationships where available; justify any model-based estimate, training evidence and evaluation. Do not replace all missing values with one blanket mean, median or mode. Retain all 500 order IDs and preserve correct observed values.

Remove outliers from the outlier batch. Use the seasonal relationship, residual diagnostics and a justified decision rule to identify anomalous delivery charges. Remove the identified rows; do not repair their charges and keep the rows. Preserve every retained row's valid values. Explain how your method avoids deleting unusual but valid orders.

Explain and reproduce your approach. Include your reasoning, assumptions, selected functions or libraries, model diagnostics, material alternatives and results in the notebook. Show the connection between each diagnosis and its repair or removal. Keep useful cell outputs visible and use concise code comments. Methods can differ from an example or template if they satisfy the published rules and are supported by evidence.

Task 1 comprises dirty outputs 6.0 marks, missing outputs 3.0, outlier outputs 2.0, methodology 3.0, and documentation and reproducibility 1.0. Correct CSVs do not automatically demonstrate a sound method; a local output error does not automatically remove all methodology marks. The documentation and reproducibility criteria assess the single notebook across all three tasks.

Task 2 Audit trail and validation (2 marks)

Generate `audit_log.csv` from the workflow. Use these columns, in this order:

Column	Required content
<code>source_file</code>	Exact incoming CSV filename
<code>record_key</code>	Affected <code>order_id</code>
<code>field_name</code>	Corrected field; use <code>delivery_charges</code> for an outlier removal
<code>issue_type</code>	<code>dirty</code> , <code>missing</code> or <code>outlier_row</code>
<code>original_value</code>	Original value; write <code><MISSING></code> for an empty source field

Column	Required content
<code>corrected_value</code>	Submitted replacement value, or <code>ROW_REMOVED</code> for a removed outlier
<code>method</code>	The rule, calculation or estimation used
<code>evidence</code>	Supporting relationship, model diagnostic or identifiable notebook evidence
<code>confidence</code>	<code>high</code> , <code>medium</code> or <code>low</code> ; define these levels in the notebook

Log one row per changed field and one row per removed outlier. Do not log every unchanged cell or harmless CSV quoting changes. The log must describe what your code actually changed, including uncertain decisions; a claim that AI approved the answer is not verification evidence. A section or check ID can provide a short, traceable reference without repeating lengthy prose in each row.

After writing the outputs, read them back and show executable checks for:

- exact schema and column order, parseable values and required completeness;
- unique, preserved order IDs; 500 dirty and 500 missing rows; a reconciled list of removed and retained outlier IDs;
- valid shopping-cart structure and identified current prices;
- calendar, warehouse, location, happiness and arithmetic relationships;
- seasonal fit and residual behaviour of retained orders;
- preservation of originally valid information; and
- agreement between input/output differences and the audit log.

Record each check's observed result, pass/fail outcome and interpretation or resolution in the notebook. Do not fabricate a value to make a check pass. A correctly detected unresolved issue should be explained and can still provide evidence of validation skill. Validation evidence earns credit independently of whether every final cell is correct.

Use the checks together: a consistency check does not replace a completeness check, and one passing check does not establish that the whole output is correct.

Task 3 Customer profiling and transformation (3 marks)

Purpose. Prepare numeric customer-profile attributes for possible future distance-based exploratory segmentation: comparing customers by their activity, spending and/or delivery geography. Investigate whether differences in units, scales or distributions would dominate those comparisons. You do not need to fit a clustering or prediction model, choose a number of clusters, or produce a dashboard. There is no single correct set of transformations.

Build a customer-level analysis copy. Read and concatenate the three cleaned CSVs you are submitting, retaining their source-batch identity. Use all retained orders, including only the rows retained in your outlier output. **Do not restore removed orders to reach 1,500 rows, append A1 orders, or use another group's data.** Report the input row count for each cleaned batch, the combined distinct order count and the number of customers. Aggregate to one row per `customer_id`. These profiles describe the retained A2 sample, not complete customer histories or lifetime value. Explain any unresolved input problems affecting the analysis.

Select **three attributes** from this candidate set. You only need to derive your selection.

Attribute	Customer-level definition over retained orders	Type
<code>order_count</code>	Count of distinct <code>order_id</code> values	Non-monetary
<code>total_paid</code>	Sum of <code>order_total</code>	Monetary, AUD
<code>mean_order_value</code>	Mean of <code>order_total</code>	Monetary, AUD
<code>mean_delivery_distance</code>	Mean of <code>distance_to_nearest_warehouse</code>	Non-monetary, km

The paid-value attributes include delivery charges after the item discount; they are not profit or merchandise-only revenue. Do not transform customer IDs. If you select count, total and mean together, discuss their algebraic dependence and the risk of counting the same spending information more than once in a distance calculation.

Compare and interpret. Show the original scales and distributions of the selected attributes and explain their implications for the stated purpose. Keep an untransformed baseline, then:

- compare standardisation (z-score) and min-max scaling on the three attributes;
- compare at least one applicable nonlinear transformation, such as log, log1p or square root, against the original values of at least one selected attribute; explain its domain requirements and suitability; and
- use compact numerical summaries and/or plots to show the actual effects, including relevant relationships between the selected attributes. State the order if combining a nonlinear transformation with scaling.

You do not need every method on every attribute, apart from the two scaling comparisons specified above. Existing notebook diagnostics may be referenced instead of repeated. Explain

handling of zeros, constant values or other domain constraints where relevant. A skewed attribute does not automatically need a nonlinear transformation. Achieving normality, reducing skewness, or increasing R-squared is **not** a universal success condition. The seasonal delivery model in Task 1 has a different purpose; Task 3 does not change its predictors or rules.

Make a recommendation. Include one concise table with a row for each selected attribute: attribute, methods compared, observed effect or evidence reference, recommended treatment, and justification/limitation. Keeping an attribute unchanged can earn full credit when supported by a meaningful comparison. In the recommended treatment, state the **nonlinear transformation (or none)** and **scaling (or none)** separately, including their order if both are recommended. For example, "no nonlinear transformation; standardisation" is different from "no nonlinear transformation; no scaling". If you recommend no scaling, explain how differences in units and scale are handled or justified for the stated purpose; low skewness alone does not answer this question.

Discuss effects on relative distances, extreme observations, interpretability and/or redundant features as relevant, without claiming that the comparison proves meaningful customer clusters exist. If you optionally evaluate a held-out model, fit transformation parameters on its training data only; a held-out model is not required for this task.

Task 3 comprises comparison and evidence (T1, 2 marks) and recommendations, boundaries and limitations (T2, 1 mark). Reasonable, evidence-based choices may differ between groups. A Task 1 error does not automatically remove Task 3 credit; your analysis must still be transparent and defensible on its inputs.

Keep the customer profiles and transformed features in the notebook analysis only. No additional CSV or report is required. **Do not replace values in the three cleaned CSVs with transformed values**, or record an analytical transformation as a cleansing correction in `audit_log.csv`.

There is **no separate EDA PDF, fixed figure count, ten-findings requirement, five-ML-questions requirement, or ten-page report limit** for this A2. Use the single solution notebook as your technical report; include the diagnostic plots and tables that support your decisions. Additional length alone earns no credit. You may organise section titles and workflow order to suit your approach, provided all three tasks and rubric evidence are easy to locate.

5 Required outputs and comparison conventions

Input	Required output	Row requirement
<code>GroupNNN_dirty_data.csv</code>	<code>GroupNNN_dirty_clean.csv</code>	All original 500 order IDs

Input	Required output	Row requirement
GroupNNN_missing_data.csv	GroupNNN_missing_clean.csv	All original 500 order IDs
GroupNNN_outlier_data.csv	GroupNNN_outlier_clean.csv	Only retained original order IDs

All three outputs use the original 16-column schema, column order and units. Task 3 scaling and transformation must not change these deliverables. Do not add audit columns or a pandas index. Write valid UTF-8 CSVs and retain identifiers as strings. Matching uses `order_id`, so row order is not assessed. Valid numeric representations such as `8` and `8.00` are equivalent; booleans may be `True/False` or `1/0` (case-insensitive for the words). Standard CSV escaping is expected. Equivalent JSON whitespace, object-key order and the order of unchanged cart entries do not change a cart's meaning; product names and quantities must be preserved.

The following absolute tolerances apply **to fields that require correction or imputation**, with a negligible floating-point comparison allowance. There is no percentage-based relative tolerance.

Field	Accepted absolute difference from the reference
<code>order_price, order_total</code>	At most AUD 0.01
<code>delivery_charges</code>	At most AUD 0.08
<code>distance_to_nearest_warehouse</code>	At most 0.0001 km
<code>customer_lat, customer_long</code>	At most 0.0000001 degrees
IDs, dates, categories and other non-numeric values	Correct value under the stated format and equivalence rules

Originally valid values must remain equal in meaning to the source; the wider model tolerance does not permit changing every delivery charge to a fitted value. Deleting correct outlier rows or introducing new errors reduces output credit as defined in the rubric.

Use the specified filenames and structure. An affected output receives zero output credit if it is missing, unreadable, has missing/extra/renamed/reordered columns, duplicate order IDs, unexpected order IDs, or (for dirty/missing) does not retain the complete input ID set. Other output files and independently demonstrated methodology remain accessible. A filename case difference or extra ZIP wrapper that still identifies the required file unambiguously does not by

itself make the output unassessable. Incorrect cell values in an otherwise readable, correctly structured CSV are evaluated by the cell-comparison rules.

6 Submission and reproducibility

Submit `GroupNNN_A2_submission.zip`, containing one folder named `GroupNNN_A2_submission` with the following files:

File	Required content
<code>GroupNNN_dirty_clean.csv</code>	Dirty-data output
<code>GroupNNN_missing_clean.csv</code>	Missing-data output
<code>GroupNNN_outlier_clean.csv</code>	Outlier-data output
<code>audit_log.csv</code>	Generated field-level change and removal log
<code>GroupNNN_solution.ipynb</code>	Executed notebook covering all three tasks, methods, diagnostics, validation, transformation comparisons and recommendations
<code>GroupNNN_solution.py</code>	Corresponding Python source; include any additional imported helper modules
<code>requirements.txt</code>	Python dependency versions; state the Python version in the notebook
<code>GroupNNN_AI_declaration.pdf</code>	Group declaration described below
<code>AI_records/</code> if applicable	Complete relevant conversational-AI exports and the English index

The notebook is the technical report. A separate HTML or PDF audit report is not required. The `.py` source should correspond to the submitted notebook; printed outputs need not be included in the `.py` file. Keep outputs in the submitted notebook, even if you use a temporary copy with cleared outputs when exporting source code.

Put input/output path settings and `GROUP_ID` in one clearly labelled configuration cell or section. Use relative or configurable paths. For example, `DATA_DIR = Path("Group007_A2")` can point to an extracted input directory beside the notebook. Explain the expected layout. A marker can configure the input and output roots; they should not need to edit your cleaning logic or recreate your personal Drive setup.

The teaching team retains the allocated input package. **Do not resubmit the raw data or lexicon**, require a private/shared-drive mount, or download data or resources during execution. Once declared dependencies are installed, the work must run offline with the supplied local resources. Restart the kernel and Run All before submission, verify that it recreates the three CSVs and audit log, and re-read those files in Python. The same run must reproduce the Task 3 customer profiles and comparison results in the notebook. Do not overwrite the original inputs.

There are no additional package restrictions, but declare dependencies and make any nondeterministic evaluation split reproducible. Prefer efficient operations and avoid unnecessary repeated file reads, downloads or model fits.

7 AI use attribution and records

Generative AI may be used throughout A2. Your group remains responsible for understanding the code, verifying its results and documenting material uses. Using AI is not a separate scored criterion; the rubric assesses the submitted data work. Reproducible checks and justified decisions are still required.

In [GroupNNN_AI_declaration.pdf](#), identify the group and all members, list tools and purposes, summarise material uses, and describe how important suggestions were independently checked. All members must sign. If no AI was used, state that explicitly. You may reuse the A1 declaration layout with the assessment identifier updated to A2; the fields in this section are the requirements.

For conversational AI, put complete A2-related conversations in [AI_records/](#) and include [GroupNNN_AI_index.pdf](#) in that folder. The index must be in English and identify the tool, member, export filename, purpose, where its output was used, and verification performed. The declaration may refer to this index; there is no need to reproduce the same table twice. Keep the original languages of conversations. Readable PDF, HTML, TXT, DOCX and native JSON exports are accepted; JSON must preserve the complete relevant conversation in order. Do not submit unrelated personal conversations or rely solely on live links. For non-conversational tools without chat exports, describe the tool, affected work and verification in the declaration.

Independent verification means checking an AI suggestion against data, a known relationship, executable tests or another appropriate source of evidence. For example, test an inferred price vector on suitable unused cart equations, or compare proposed outlier decisions with residual diagnostics and retained-row checks. Asking the same AI whether its answer is correct is not sufficient.

Reference external code and sources. Hard-coded row-specific repairs, private answer lists or manual output editing do not demonstrate a reproducible cleansing method. Declaration or integrity concerns are handled through the applicable assessment process, not an additional unpublished mark deduction.

Appendix Submission checklist

- The ZIP and all group-specific filenames use your allocated three-digit ID.
- The three corrected CSVs have the correct columns, types, IDs and row rules.
- You have checked repairs, removed rows and preservation of valid values.
- The audit log agrees with actual output changes and identifies the evidence.
- The executed notebook explains all three tasks and retains useful results.
- Task 3 states the retained sample and grain, compares the required methods on three attributes, and includes a justified recommendation table.
- Matching Python source, any imported helpers and dependency information are included.
- A fresh local Run All recreates the submitted outputs from configurable paths.
- The signed AI declaration and applicable records/index are included.
- Original inputs, local environments and temporary files are excluded.
- The ZIP can be extracted and its CSVs read by pandas. Confirm the Moodle submission activity shows your group's submission as complete.